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Author(s): Otis Dudley Duncan and Beverly Davis

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AN ALTERNATIVE TO ECOLOGICAL CORRELATION

OTIS DUDLEY DUNCAN AND BEVERLY DAVIS

University of Chicago

It was shown by Robinson that ecological correlation may be seriously misleading if it is used to infer corresponding individual correlation.¹ Herbert Menzel's "Comment" on Robinson's paper made the valid observation that "ecological correlations may be of great value even without reflecting individual correlations."² We make the additional point that, even if the investigator is concerned with the individual correlation, data classified by area may be of service, though ecological correlation is probably not the best way to handle them.³

Under favorable conditions, the individual correlation may be determined to a fair approximation, using principles set forth by Yule for testing consistence of data and for drawing inferences from incomplete data.⁴ This proposition is illustrated below with census tract data for Chicago taken from the publications of the 16th and 17th decennial censuses.

Table 1 shows the cross tabulation of occupation by color for the employed females of the city in 1940. Six per cent of the whites and 38 per cent of the nonwhites are employed in domestic service; or expressing these same facts in another form, the fourfold-point correlation between occupation and color is .289. In the situation where ecological correlation has been used in the past, and for which we are proposing an alternative technique, only the marginal totals of Table 1 are available, but there is also available a similar set of marginal totals for each of the areal subdivisions—say tracts—of the universe under study. Given just the marginals of Table 1, we may infer only that the percentage of nonwhites engaged in domestic service is between zero and 100, and that the fourfold r may be as low as $-.008$ and as high as $.898$. In tract 625 there are 877 employed females of whom 95 are white, 782 nonwhite; 239 are in domestic service and 638 in other occupations. At a maximum, therefore, only 239

nonwhites are in domestic service, while at least 144 (239-95) nonwhites must be in this occupation. Making a similar computation for each of the 935 tracts of the city and summing the maximum and minimum frequencies, we find that between 25 and 40 per cent of the nonwhites must be engaged in domestic service, and accordingly the fourfold r is between .126 and .355. (Note that it is necessary to fix maximum and minimum frequencies for only one cell of Table 1, since this cell, along with the marginals, determines the remaining frequencies.)

TABLE 1. EMPLOYED FEMALES BY OCCUPATION AND COLOR FOR CHICAGO, 1940

Color	Occupation		
	Other than Domestic Service	Domestic Service	All Occupations
White	359,011	23,241	382,252
Nonwhite	17,168	10,433	27,601
Total	376,179	33,674	409,853

An even closer approximation to the individual correlation is obtained using tract data for color of occupants and tenure of occupied dwelling units. We have the following marginals for the 949,744 occupied d.u.'s in the city in 1940: owner occupied, 230,975; tenant occupied, 718,769; occupied by whites, 873,479; occupied by nonwhites, 76,265. The city marginals are consistent with any fourfold r between $-.521$ and $.168$, whereas this range is narrowed to $.058$ to $.138$ on the basis of the tract marginals, as compared to the actual r of $.116$.

A third example is more realistic than the foregoing, in the sense that the city cross tabulation is not available. In 1950 the 1,106,119 d.u.'s of the city were distributed by year built and persons per room as follows: built later than 1929, 89,170; built in 1929 or earlier (including not reported), 1,016,949; less than 1.01 persons per room (including unoccupied and not reported), 943,112; 1.01 or more persons per room, 163,007. The city marginals permit variation between zero and 100 in the percentage of newer d.u.'s which are overcrowded, while this range is narrowed to 1.4-44.2 per cent with the tract data. Similarly the fourfold r is determined to be between $-.246$ and $.111$ from the tract data, although the wider limits of $-.712$ and $.123$ are consistent with the city marginals.

Table 2 summarizes the three illustrations. It will be noted that substantially closer ap-

¹ W. S. Robinson, "Ecological Correlations and the Behavior of Individuals," *American Sociological Review*, XV (June, 1950), pp. 351-357.

² *Ibid.*, p. 674.

³ We believe there are superior alternatives to correlation for strictly ecological studies as well; but that is the subject for another discussion.

⁴ G. Udny Yule and M. G. Kendall, *An Introduction to the Theory of Statistics*, 13th ed., London: Griffin and Co., 1947, Ch. 2.

TABLE 2. ILLUSTRATIONS OF INDIVIDUAL CORRELATIONS APPROXIMATED WITH AREA DATA

Variables Correlated and Parameters Computed	Basis of Calculation			
	City Cross Tabulation	City Marginals	City and Community Area Marginals	City and Tract Marginals
1. Color and occupation of employed females, 1940				
Percentage of non- whites in domestic service	37.8	0.0—100.0	21.1—44.5	25.1—40.7
Fourfold r	.289	— .008—.898	.126—.355	.165—.317
2. Tenure and color of occupants of occupied dwelling units, 1940				
Percentage of non- white occupied d.u.'s tenant occupied	92.5	0.0—100.0	75.4—98.0	84.1—95.8
Fourfold r	.116	— .521—.168	— .002—.153	.058—.138
3. Year built and persons per room of dwelling units, 1950				
Percentage of d.u.'s built since 1929 hav- ing 1.01+ per per- sons per room		0.0—100.0		1.4—44.2
Fourfold r		— .712—.123		— .246—.111

proximations to the individual correlation can be made from the marginal frequencies for the 935 census tracts than from the marginals for the 75 community areas, which are combinations of tracts. Although different systems of areal subdivision give different results, as is the case with ecological correlation,⁵ the criterion for the choice among these results is clear. The individual correlation is approximated most closely by the least maximum and the greatest minimum among the results for several systems of areal subdivision.

It should be noted that the range of possibilities determined from the tract marginals is not in the nature of a set of confidence limits for the true frequency or true correlation, but rather provides absolute upper and lower bounds for their values. Further, such a range need not be symmetrical about the actual value.

It is not suggested that the illustrations cited in this paper are typical. Furthermore, the investigator must decide in each case whether the approximation to the individual correlation is close enough for his purposes. However, the suggested procedure should probably be tried before adopting the unsatisfactory expedient of accepting the ecological correlation as an estimate of the individual correlation.

⁵ Robinson, *op. cit.*, p. 356

OPPORTUNITIES FOR SOCIOLOGICAL RESEARCH IN ALASKA

MARGARET LANTIS

*Arctic—Desert—Tropic Information Center,
U. S. Air Force*

Sociologists do not seem to know that they are missing a chance at very worthwhile research, and pleasant travel to get to it. To draw attention to the Territory's need for social research, the Alaska Division of the American Association for the Advancement of Science took special measures at the Third Alaskan Science Conference, held in September 1952, at which meeting the current social, political, and economic status of Alaska (chiefly the last two) and the status of social research were discussed.¹ This was planned without a single sociologist, because there were none in Alaska. At the University of Alaska no courses in sociology were being given except two offered apologetically by an anthropologist. President Terris Moore wanted to give more, but his school was small and he had many simultaneous

¹ The three papers given at a first-morning general session of the Third Alaskan Science Conference have been published in *The Scientific Monthly*, July 1953.